

Empirical Evaluation of Predictive Models for Audio-Feature-Based Track Popularity

Group Members

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Problem Statement

Anticipating user engagement with audio tracks requires mapping high-dimensional, non-linear acoustic features to a continuous engagement metric. Existing heuristics often fail to capture the complex interplay between low-level audio properties and broader listening trends, particularly given the zero-inflated nature of popularity scores where a vast majority of tracks receive minimal engagement. This project investigates the efficacy of various machine learning architectures in modeling this relationship to predict track popularity, providing a robust framework for understanding feature importance in audio-centric environments.

General Approach

We formulate popularity prediction as a supervised regression task over tabular acoustic features. We will conduct a comparative analysis, benchmarking classical linear and tree-based ensemble methods against deep feed-forward architectures implemented in PyTorch. Key technical challenges include mitigating the effects of skewed target distributions (popularity bias) and optimizing hyperparameter configurations to prevent overfitting on noisy, real-world target variables.

Data

We will utilize the Kaggle "Spotify Tracks Dataset," a pre-compiled corpus of approximately 114,000 instances. The dataset provides continuous and categorical features derived from Spotify's audio analysis API (e.g., acousticness, valence, instrumentalness, tempo) alongside a normalized ground-truth popularity score (0–100). Relying on this structured dataset abstracts away API scraping overhead, allowing us to focus our computational resources entirely on feature normalization, model optimization, and rigorous evaluation.

Dataset Link: <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

Measure of Success

Empirical success will be quantified by minimizing generalization error on a held-out test set, primarily evaluated via Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). Specifically, we define predictive success as achieving an R^2 score of ≥ 0.50 and demonstrating at least a 15% reduction in RMSE compared to a naive baseline model (e.g., predicting the mean popularity). Furthermore, the project's broader success relies on a rigorous comparative study, clearly demonstrating the trade-offs in computational cost, interpretability, and predictive performance between traditional statistical models and deep neural networks on structured audio data.

Milestones and Deliverables

- **Milestone 1 (Week1):** Dataset acquisition, exploratory data analysis (EDA), feature scaling/normalization, and construction of PyTorch DataLoader pipelines.
- **Milestone 2 (Week2):** Implementation, cross-validation, and hyperparameter tuning of baseline regressors (e.g., Ridge/Lasso Regression, Random Forests).
- **Milestone 3 (Week3):** Architecture design, implementation, and training of the deep neural network using PyTorch, including optimization of learning rates and regularization parameters (e.g., weight decay, dropout).
- **Milestone 4 (Week4):** Final empirical evaluation, generation of learning curves and feature importance plots, and submission of the final project report along with the reproducible code repository.